

4.3.1 Keys and Integrity Constraints

کینڈیڈیٹ کی وہ فیلڈ ہے جو ریکارڈ کی شناخت کر سکے۔

CS XI

Chapter 4: Data and Analysis

Lecture 04 of 15

★ Golden

4.3.1 Keys and Integrity Constraints ★ Golden

LEARN IT

A **key** is a field (or a set of fields) that uniquely identifies records and links tables. Integrity constraints are the rules that keep those keys honest.

Candidate key. Any field (or combination) that *could* uniquely identify a row. A table may have several. In an Employee table: EmpID, EmpLicense# and EmpPassport are all candidate keys.

Primary key (PK). The one candidate you actually choose. Rules: unique, never NULL, only one PK per table (it may be a **composite** of two columns). Example: EmpID.

Alternate key. The leftover candidate keys. After EmpID is chosen, EmpLicense# and EmpPassport become alternate keys — still unique, just not “the” identifier.

Foreign key (FK). A field in this table that refers to the PK of another table. It *may* contain duplicates (many employees in one department) and, in some designs, NULL.

WORKED EXAMPLE

Classroom analogy. Roll Number, B-Form number and phone number can all identify a student (candidate keys). The school picks Roll Number as the primary key. B-Form and phone become alternate keys. In a **Result** table, RollNo is a foreign key pointing back to Student.

Table	Field	Role
Student	RollNo	Primary key
Student	BFormNo	Alternate key
Result	RollNo	Foreign key → Student.RollNo
Result	ExamID + RollNo	Could be a composite primary key

EXAM TIP

Board questions: "Differentiate primary and foreign key" (table of 4 points) and "Why can a PK not be NULL?" Because a NULL would mean "this row has no identity", so the DBMS could not tell rows apart.

COMMON MISTAKE

A phone number is a bad PK: people change numbers, two siblings may share one, and some students have none (NULL). Roll numbers are issued by the school and stay unique.

اردو میں سمجھیں

کینڈیڈیٹ کی وہ فیلڈ ہے جو ریکارڈ کی شناخت کر سکے۔ پرائمری کی منتخب شدہ شناخت ہے — منفرد اور خالی نہیں۔ جو بچ جائیں وہ آلٹرنیٹ کیز ہیں۔ فارن کی دوسرے ٹیبل کی پرائمری کی طرف اشارہ کرتی ہے اور ٹیبلز کو جوڑتی ہے۔

CHECK YOURSELF

1. Can a table have two primary keys?
2. Can a foreign key have duplicate values?

Answers: 1. No — only one PK (it may be composite). 2. Yes — many child rows may point to the same parent.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. A primary key:

- a. Uniquely identifies each record
- b. May freely repeat
- c. Is always a name
- d. Is a chart

MCQ 2. A candidate key is:

- a. Any field set that could be the PK
- b. A foreign key
- c. A report
- d. A form

MCQ 3. An alternate key is:

- a. A candidate not chosen as PK
- b. A duplicate PK
- c. A null PK
- d. A pie

MCQ 4. A foreign key:

- a. Refers to a PK in another table
- b. Is always unique in its table
- c. Sorts the table
- d. Encrypts the DB

MCQ 5. NULL in a primary key is:

- a. Not allowed
- b. Required
- c. A default
- d. A relationship type

MCQ 6. Integrity constraints exist to:

- a. Stop invalid data
- b. Draw graphs
- c. Route packets
- d. Compile Python

Extra short questions

1. Define PK, FK, candidate, alternate.
2. Give a two-table example with an FK.
3. Entity integrity rule.
4. Why not use student name as PK?

Extra long question

1. For Members and Borrow in a library, choose keys and write two integrity rules. Show one invalid insert the DBMS should reject. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. PK: chosen unique identifier. Candidate: any unique identifier. Alternate: unused candidate. FK: copy of another table's PK.
2. Students(roll PK); Marks(roll FK, subject, score).
3. PK unique and not NULL.
4. Names duplicate and change.

Long

1. MemberID PK; Borrow(BorrowID PK, MemberID FK). Reject Borrow with unknown MemberID or NULL MemberID. Mention cascade later.

Next lecture: 4.3 Relational Database Management System (RDBMS)